Final Examination Practice Sheet - ANSWER KEY

Course Code: CSE563
Course Title: APPLIED MATHEMATICS
Section(s): K3P25RB
Syllabus Covered: Unit 1 to Unit 6
Pattern: per unit, 8 MCQs (1 mark) and 2 long questions (10 marks)
Purpose: self-practice for the end-term examination

For the instructor. Every numerical answer in this key was recomputed and verified before printing. The Source column names the syllabus topic each question is drawn from.

Unit 1: Basic Mathematics in Computer Science

Part A - MCQ answers

Q	Answer	Reason / working	Source
1	(C) 12	Euclid: 180 = 3(48) + 36; 48 = 1(36) + 12; 36 = 3(12) + 0, so the gcd is 12.	Number theory
2	(A) 64	A set with n elements has 2^n subsets; $2^6 = 64$.	Set theory
3	(D) 1	11 is prime and does not divide 2, so $2^{10} = 2^{p-1} \equiv 1 \pmod{11}$.	Number theory
4	(B) if not q then not p	The contrapositive negates and swaps both parts, and it is the only one of the three derived forms that is logically equivalent to the original.	Logic
5	(B) 42	Inclusion-exclusion: 30 + 20 − 8 = 42.	Set theory
6	(D) 2xy	Differentiate with respect to x treating y as a constant: $\partial(x^2y)/\partial x = 2xy$, and $3y^2$ contributes nothing.	Calculus
7	(A) reflexivity and symmetry	a < a is false, so it is not reflexive; a < b does not give b < a, so it is not symmetric. Transitivity does hold.	Relations
8	(C) 8	One minterm for each of the $2^3 = 8$ input combinations.	Boolean algebra

Part B - Expected answers and marking scheme

9. Apply the Euclidean and extended Euclidean algorithms to 1071 and 462. (10 Marks)

(a) State the division algorithm, and state the exact condition under which a has a multiplicative inverse modulo m.

Division algorithm: for any integer a and positive integer d there exist unique integers q and r with $a = dq + r$ and $0 \le r < d$. An inverse of a modulo m exists if and only if gcd(a, m) = 1. **(3 marks.)**

(b) Find gcd(1071, 462) by the Euclidean algorithm, then express that gcd in the form 1071x + 462y.

1071 = 2(462) + 147; 462 = 3(147) + 21; 147 = 7(21) + 0, so the gcd is 21. Back-substituting: 21 = 462 − 3(147) = 462 − 3(1071 − 2(462)) = 7(462) − 3(1071). So x = −3 and y = 7. **(4 marks.)**

(c) Verify your identity by substitution, and state the lcm of the two numbers.

−3(1071) + 7(462) = −3213 + 3234 = 21. Correct. lcm = (1071 × 462) / 21 = 494802 / 21 = 23562. **(3 marks.)**

10. Explain and apply proof by mathematical induction. (10 Marks)

(a) State the basis step and the inductive step, and explain why a proof that omits either one is worthless.

Basis: show the statement holds for the first value. Inductive step: assume it holds for an arbitrary k and prove it for k + 1. Without the basis the chain has nothing to start from, so a false statement can satisfy the inductive step alone. Without the inductive step only one case has been checked. **(4 marks.)**

(b) Prove by induction that 3 divides $n^3 - n$ for every integer $n \ge 0$.

Basis (n = 0): $0^3 - 0 = 0 = 3(0)$. Hypothesis: $k^3 - k = 3m$ for some integer m. Step: $(k+1)^3 - (k+1) = k^3 + 3k^2 + 3k + 1 - k - 1 = (k^3 - k) + 3k^2 + 3k = 3m + 3(k^2 + k) = 3(m + k^2 + k)$, which is divisible by 3. **(4 marks.)**

(c) Give the one-line alternative argument that avoids induction entirely.

$n^3 - n = (n - 1)n(n + 1)$, a product of three consecutive integers, one of which must be a multiple of 3. **(2 marks.)**

Unit 2: Vectors and Matrices

Part A - MCQ answers

Q	Answer	Reason / working	Source
1	(B) −14	$3(6) - 8(4) = 18 - 32 = -14$.	Determinants
2	(D) 12	$1(4) + 2(-5) + 3(6) = 4 - 10 + 18 = 12$.	Vectors
3	(A) 10	$\sqrt{(36 + 64)} = \sqrt{100} = 10$.	Vectors
4	(C) because a 2×2 linear map always fixes the origin and cannot add an offset	Translation is affine, not linear. A 2×2 matrix M satisfies $M0 = 0$, so it can never shift the origin; the extra coordinate turns the offset into ordinary matrix multiplication.	Image transformation
5	(C) 5 and 3	For a diagonal matrix the eigenvalues are the diagonal entries themselves.	Eigenvalues
6	(A) subtract the mean of each feature	PCA measures variance about the mean, so the data must be centred first; otherwise the first component merely points at the mean.	PCA
7	(D) symmetric	$S = X^TX/(n-1)$ satisfies $S^T = S$, which by the spectral theorem gives real eigenvalues and orthogonal eigenvectors.	PCA
8	(B) nearest neighbour	Mask values are class identifiers, not intensities: averaging label 2 and label 6 to get 4 invents a class. Nearest neighbour returns only values that exist in the source.	Image interpolation

Part B - Expected answers and marking scheme

9. Apply Principal Component Analysis by hand to the five points (2, 1), (3, 3), (4, 3), (5, 5) and (6, 8). (10 Marks)

(a) Explain why the data must be centred, and state what the covariance matrix captures.

PCA looks for the directions of greatest variance, and variance is measured about the mean, so the mean must first be moved to the origin. The covariance matrix holds the variance of each feature on its diagonal and the covariance between features off the diagonal. **(3 marks.)**

(b) Centre the data and compute the 2×2 covariance matrix.

Means: $\bar{x} = 20/5 = 4$, $\bar{y} = 20/5 = 4$. Centred: (−2, −3), (−1, −1), (0, −1), (1, 1), (2, 4). Sums: $\Sigma x^2 = 10$, $\Sigma y^2 = 28$, $\Sigma xy = 16$. Dividing by $n - 1 = 4$ gives $S = [[2.5, 4], [4, 7]]$. **(4 marks.)**

(c) Find both eigenvalues of S and state the percentage of variance carried by the first principal component.

$\lambda^2 - 9.5\lambda + 1.5 = 0$, so $\lambda = (9.5 \pm \sqrt{84.25})/2$, giving $\lambda_1 = 9.3394$ and $\lambda_2 = 0.1606$ (their sum is 9.5, the trace, as a check). PC1 carries $9.3394/9.5 = 98.3$ per cent of the variance, so the data are very nearly one-dimensional. **(3 marks.)**

10. Apply homogeneous coordinates to rotate the point (4, 2) by 90 degrees anticlockwise about the point (2, 2). (10 Marks)

(a) Explain why the rotation cannot be done with a single 2×2 rotation matrix, and give the sequence of transformations needed.

A 2×2 rotation matrix always rotates about the origin. To rotate about another centre the sequence is: translate the centre to the origin, rotate, then translate back, i.e. $M = T(2, 2)\, R(90°)\, T(-2, -2)$. **(3 marks.)**

(b) Write the three 3×3 matrices and multiply them into a single matrix M.

With $\cos 90° = 0$ and $\sin 90° = 1$, $M = [[1,0,2],[0,1,2],[0,0,1]] \cdot [[0,-1,0],[1,0,0],[0,0,1]] \cdot [[1,0,-2],[0,1,-2],[0,0,1]] = [[0, -1, 4], [1, 0, 0], [0, 0, 1]]$. **(4 marks.)**

(c) Apply M to (4, 2) and verify the result geometrically.

$x' = 0(4) - 1(2) + 4 = 2$; $y' = 1(4) + 0(2) + 0 = 4$, giving (2, 4). Check: relative to the centre the point is at offset (2, 0), two units due east; a 90° anticlockwise turn sends east to north, giving offset (0, 2), i.e. the point (2, 4). The distance from the centre is 2 before and after, as a rotation requires. **(3 marks.)**

Unit 3: Probability and Statistics

Part A - MCQ answers

Q	Answer	Reason / working	Source
1	(D) 1/9	The favourable rolls are (3,6), (4,5), (5,4), (6,3): 4 out of 36, which is 1/9.	Basic probability
2	(B) 6	$E[X] = np = 20(0.3) = 6$.	Distributions
3	(C) 0.24	$Var = p(1 - p) = 0.4(0.6) = 0.24$.	Distributions
4	(A) 68 per cent	The empirical rule: about 68 per cent of the distribution lies within one standard deviation of the mean.	Distributions
5	(A) 81 per cent of the	R^2 is the proportion of the variance in the response explained by the	Regression

	variance in y is explained by the model	model. Here the correlation would be $\sqrt{0.81} = 0.9$, not 0.81.	
6	(C) rejecting a true null hypothesis	Type I is a false positive: the null hypothesis was true and was rejected anyway. Type II is the reverse.	Inference
7	(B) 3.5	$(1 + 2 + 3 + 4 + 5 + 6)/6 = 21/6 = 3.5$.	Distributions
8	(D) prior	Prior × likelihood, divided by the evidence, gives the posterior.	Bayesian inference

Part B - Expected answers and marking scheme

9. Apply Bayes' theorem to a screening test. A disease affects 2 per cent of a population; the test is 95 per cent sensitive and 90 per cent specific. (10 Marks)

(a) State Bayes' theorem, and define prior, likelihood and posterior.

$P(D \mid +) = P(+ \mid D)\,P(D) / P(+)$. The prior $P(D)$ is the probability of the disease before testing; the likelihood $P(+ \mid D)$ is the probability of the observed result if the person has the disease; the posterior $P(D \mid +)$ is the revised probability after seeing the result. **(3 marks.)**

(b) Find the probability that a person who tests positive actually has the disease, showing all working.

$P(D) = 0.02$, $P(+ \mid D) = 0.95$, $P(+ \mid \text{not } D) = 1 - 0.90 = 0.10$. $P(+) = 0.02(0.95) + 0.98(0.10) = 0.019 + 0.098 = 0.117$. $P(D \mid +) = 0.019 / 0.117 = 0.1624$, that is 16.24 per cent. **(4 marks.)**

(c) Explain why that figure is so far below 95 per cent.

Because the disease is rare. The 98 per cent who are healthy generate 0.098 false positives, more than five times the 0.019 true positives from the 2 per cent who are ill, so most positive results are false. Sensitivity says nothing on its own; the base rate dominates. **(3 marks.)**

10. Fit and interpret a least-squares regression line for the pairs (1, 2), (2, 4), (3, 5), (4, 4) and (5, 5). (10 Marks)

(a) State what least squares minimises, and explain why the vertical distances are squared rather than simply added.

It minimises the sum of the squared vertical distances between the observed and fitted values. Squaring prevents positive and negative residuals from cancelling, and it penalises large misses disproportionately, giving a single differentiable criterion with a closed-form solution. **(3 marks.)**

(b) Compute the regression line $y = a + bx$, showing the two sums you use.

$\bar{x} = 3$, $\bar{y} = 4$. Deviations in x: $-2, -1, 0, 1, 2$; in y: $-2, 0, 1, 0, 1$. $\Sigma(dx)(dy) = 4 + 0 + 0 + 0 + 2 = 6$ and $\Sigma(dx)^2 = 4 + 1 + 0 + 1 + 4 = 10$, so $b = 6/10 = 0.6$ and $a = 4 - 0.6(3) = 2.2$. The line is $y = 2.2 + 0.6x$. **(4 marks.)**

(c) Compute R^2, interpret it, and state one limitation of predicting y at $x = 6$.

$\Sigma(dy)^2 = 4 + 0 + 1 + 0 + 1 = 6$, so $r = 6/\sqrt{(10 \times 6)} = 0.7746$ and $R^2 = 0.6$: the line explains 60 per cent of the variation in y, leaving 40 per cent unexplained. At $x = 6$ the prediction is $2.2 + 0.6(6) = 5.8$, but $x = 6$ lies outside the observed range 1 to 5, so this is extrapolation and the fitted relationship is not evidence that the line continues to hold there. **(3 marks.)**

Unit 4: Graph Theory

Part A - MCQ answers

Q	Answer	Reason / working	Source
1	(A) 24	The handshaking lemma: the degrees sum to twice the number of edges, $2(12) = 24$.	Graph basics
2	(C) 15	$n(n - 1)/2 = 6(5)/2 = 15$.	Graph types
3	(B) 9	A tree on n vertices always has exactly $n - 1$ edges.	Trees
4	(D) 2	The two parts give a proper 2-colouring, and at least one edge rules out 1.	Graph colouring
5	(D) Dijkstra	Dijkstra finalises a vertex as soon as it is reached most cheaply, which a later negative edge can invalidate. Bellman–Ford and Floyd–Warshall handle negative weights.	Shortest paths
6	(B) a directed acyclic graph	A cycle would require every vertex on it to come before itself, so a topological order exists exactly for DAGs.	Traversal
7	(C) the capacity of the minimum cut	Every cut bounds the flow from above, and the theorem says the bound from the smallest cut is achieved exactly.	Max-flow min-cut
8	(A) every vertex has even degree	Each visit to a vertex uses one edge in and one out, so every degree must be even. Exactly two odd vertices gives an Eulerian path, not a	Eulerian graphs

		circuit.	

Part B - Expected answers and marking scheme

9. Apply Dijkstra's algorithm to the weighted graph with vertices A, B, C, D, E and edges AB = 4, AC = 2, BC = 1, BD = 5, CD = 8, CE = 10, DE = 2. (10 Marks)

(a) Explain the central idea of Dijkstra's algorithm, and state why it fails on graphs with negative edge weights.

It repeatedly takes the unvisited vertex with the smallest tentative distance, fixes that distance as final, and relaxes its outgoing edges. The correctness argument assumes no later path can be cheaper, which requires that adding an edge never reduces a distance; a negative edge breaks that assumption, so a vertex may be finalised too early. **(3 marks.)**

(b) Find the shortest distance from A to every other vertex, showing the path in each case.

A–C = 2 (direct). A–B = 3 via A–C–B (2 + 1), which beats the direct edge AB = 4. A–D = 8 via A–C–B–D (2 + 1 + 5), which beats A–C–D (2 + 8 = 10). A–E = 10 via A–C–B–D–E (8 + 2), which beats A–C–E (2 + 10 = 12). **(4 marks.)**

(c) Comment on what this result shows about using the direct edge between two vertices.

In three of the four cases the shortest route is not the direct edge: A to B is cheaper through C, and both D and E are reached through the A–C–B chain. A single edge being present says nothing about whether it lies on a shortest path. **(3 marks.)**

10. Apply Kruskal's algorithm to find a minimum spanning tree of the same graph (AB = 4, AC = 2, BC = 1, BD = 5, CD = 8, CE = 10, DE = 2). (10 Marks)

(a) State what a minimum spanning tree is, and describe how Kruskal's algorithm builds one.

An MST is an acyclic connected subgraph that includes every vertex and has the least possible total edge weight. Kruskal sorts all edges by weight and adds each in turn, skipping any edge whose endpoints are already connected, until n − 1 edges have been chosen. **(3 marks.)**

(b) Carry out the algorithm, listing every edge you accept and every edge you reject, and give the total weight.

Sorted: BC(1), AC(2), DE(2), AB(4), BD(5), CD(8), CE(10). Accept BC(1); accept AC(2); accept DE(2); reject AB(4), since A and B are already joined through C; accept BD(5), which links {A, B, C} to {D, E}. Four edges chosen for five vertices, so stop; CD and CE are not examined. Total weight = 1 + 2 + 2 + 5 = 10. **(4 marks.)**

(c) State how many edges an MST on n vertices must contain, and explain why the algorithm may safely reject an edge.

Exactly n − 1, here 4. An edge may be rejected because its endpoints are already connected by cheaper edges, so adding it would create a cycle without connecting anything new, and removing the heaviest edge of any cycle never disconnects the graph. **(3 marks.)**

Unit 5: Algorithm Complexity Analysis

Part A - MCQ answers

Q	Answer	Reason / working	Source
1	(C) O(log n)	Each comparison halves the remaining range, so the number of steps is about $\log_2 n$.	Searching
2	(D) Θ(n log n)	a = 2, b = 2, so n^(log_b a) = n, which matches f(n) = n. This is case 2, giving Θ(n log n) — the merge-sort recurrence.	Master theorem
3	(A) Ω(n log n)	A decision tree with n! leaves has height at least $\log_2(n!) = \Omega(n \log n)$.	Sorting
4	(B) 3-SAT	3-SAT is NP-complete; 2-SAT is solvable in polynomial time, and the other two are in P.	P–NP
5	(B) both an upper and a lower bound	Θ is a tight bound: f is squeezed between $c_1 g$ and $c_2 g$ for large n.	Asymptotic notation
6	(A) Θ(n)	Unrolling gives n terms each costing 1, so T(n) = n.	Recurrences
7	(D) O(n log n)	Merge sort splits in half and merges in linear time at every level, giving O(n log n) in the worst case as well as the best.	Sorting
8	(C) also in NP	A solution that can be found in polynomial time can certainly be checked in polynomial time, so P is contained in NP.	P–NP

Part B - Expected answers and marking scheme

9. Explain asymptotic notation and apply the Master theorem. (10 Marks)

(a) Define O, Ω and Θ, and state what each one asserts about a running time.

f(n) = O(g(n)) if f(n) ≤ c·g(n) for all n beyond some n_0: an upper bound. f(n) = Ω(g(n)) if f(n) ≥ c·g(n) similarly: a lower bound. f(n) = Θ(g(n)) if both hold, with possibly different constants: a tight bound. **(3 marks.)**

(b) Solve T(n) = 2T(n/2) + n and T(n) = 3T(n/2) + n² using the Master theorem, naming the case used in each.

First: a = 2, b = 2, $\log_2 2 = 1$, so n^(log_b a) = n = f(n). Case 2 applies, giving Θ(n log n). Second: a = 3, b = 2, $\log_2 3 = 1.585$, so n^1.585 grows more slowly than f(n) = n². Case 3 applies, and the regularity condition holds since $3(n/2)^2 = 0.75n^2 \leq c \cdot n^2$ with c = 0.75 < 1, giving Θ(n²). **(4 marks.)**

(c) Give one recurrence to which the Master theorem does not apply, and say why.

T(n) = 2T(n/2) + n/log n. Here f(n) = n/log n is smaller than n^(log_b a) = n but not smaller by any polynomial factor n^ε, so it falls in the gap between cases 1 and 2 and the theorem gives no answer. (T(n) = T(n − 1) + 1 is also outside its scope, since it does not divide the input by a constant factor.) **(3 marks.)**

10. Discuss the complexity classes P, NP, NP-hard and NP-complete. (10 Marks)

(a) Define all four classes.

P: decision problems solvable in polynomial time. NP: decision problems whose "yes" answers can be verified in polynomial time given a certificate. NP-hard: problems at least as hard as every problem in NP, to which every NP problem reduces in polynomial time; these need not be in NP or even be decision problems. NP-complete: in NP and NP-hard, the hardest problems inside NP. **(4 marks.)**

(b) Explain the role of a polynomial-time reduction, and state what the Cook–Levin theorem established.

A reduction from problem X to problem Y transforms any instance of X into an instance of Y in polynomial time, so an efficient algorithm for Y gives one for X; this transfers hardness in the opposite direction. Cook–Levin established that SAT is NP-complete, giving the first such problem and so a starting point from which every later NP-completeness proof could reduce. **(4 marks.)**

(c) State whether 2-SAT is in P and whether 3-SAT is NP-complete, with one sentence on why the difference matters.

2-SAT is in P (it reduces to checking strongly connected components of an implication graph) and 3-SAT is NP-complete. The jump from two literals per clause to three moves the problem from tractable to apparently intractable, which shows how little structural change is needed to cross the boundary. **(2 marks.)**

Unit 6: Algorithmic and Optimization Approaches

Part A - MCQ answers

Q	Answer	Reason / working	Source
1	(D) a greedy choice on the value-to-weight ratio	Because items can be split, taking as much as possible of the best ratio first is provably optimal.	Greedy
2	(A) dynamic programming	The greedy ratio rule can be beaten once items are indivisible, so the subproblem table of dynamic programming is needed.	Dynamic programming
3	(C) both optimal substructure and overlapping subproblems	Optimal substructure alone suits divide and conquer; it is the overlap that makes storing the subproblem answers worthwhile.	Dynamic programming
4	(B) greedy	It repeatedly merges the two least frequent symbols, a locally optimal choice that is provably globally optimal here.	Greedy
5	(C) divide and conquer	It splits the input in half, solves each half independently, and combines the results — with no overlapping subproblems.	Divide and conquer
6	(B) backtracking	Partial placements are extended one queen at a time and abandoned as soon as a conflict appears, which prunes most of the search tree.	Backtracking
7	(A) accepts fewer worsening moves	The acceptance probability exp(−Δ/T) shrinks as T falls, so the search shifts from exploration towards exploitation and ends close to hill climbing.	Nature-inspired methods
8	(D) no single algorithm performs best on every problem	Averaged over all possible objective functions every algorithm performs the same, so superiority must be argued for a particular class of problems rather than in general.	Nature-inspired methods

Part B - Expected answers and marking scheme

9. Compare the greedy and dynamic-programming approaches on the knapsack problem. Three items have (value, weight) = (60, 10), (100, 20) and (120, 30), and the capacity is 50. (10 Marks)

(a) State the greedy-choice property and optimal substructure, and explain which of the two knapsack versions has each.

Greedy-choice property: a globally optimal solution can be reached by making the locally best choice at each step. Optimal substructure: an optimal solution contains optimal solutions to its subproblems. The fractional version has both; the 0/1 version has optimal substructure but not the greedy-choice property. **(3 marks.)**

(b) Solve the fractional version by the greedy ratio rule, and state the true optimum of the 0/1 version.

Ratios: 60/10 = 6, 100/20 = 5, 120/30 = 4. Greedy takes all of item 1 (value 60, 10 used), all of item 2 (value 100, 30 used), then 20 of the remaining 30 units of item 3 for (20/30)(120) = 80. Fractional optimum = 60 + 100 + 80 = 240. For 0/1 the best feasible set is items 2 and 3: weight 20 + 30 = 50 exactly, value 100 + 120 = 220. **(4 marks.)**

(c) Apply the greedy rule to the 0/1 version, and explain the gap in your answers.

Greedy takes item 1 (10 used) and item 2 (30 used), leaving 20 units, which cannot hold item 3 at weight 30; it stops at value 160. That is 60 below the true optimum of 220. The best ratio commits capacity to item 1, whose small size cannot be recovered later — with indivisible items a locally best choice can block a better combination, which is exactly why the subproblem table is needed. **(3 marks.)**

10. Explain simulated annealing and the exploration-exploitation trade-off. (10 Marks)

(a) Explain what exploration and exploitation mean, and the role temperature plays in balancing them.

Exploration means sampling widely, including worse solutions, to escape a local optimum; exploitation means refining the best solution found so far. Simulated annealing starts at a high temperature, where worsening moves are accepted readily and the search explores, then cools so that such moves become progressively unlikely and the search exploits. The cooling schedule is therefore the main design choice. **(4 marks.)**

(b) The current solution has cost 50 and a candidate move has cost 54, with T = 10. Compute the probability that the move is accepted.

$\Delta = 54 - 50 = 4$, which is a worsening move, so the Metropolis rule gives $P = \exp(-\Delta/T) = \exp(-4/10) = \exp(-0.4) = 0.6703$, about 67 per cent. (A move that improved the cost would be accepted with probability 1.) **(3 marks.)**

(c) State what the algorithm becomes as T approaches 0, and name the risk this creates.

As $T \to 0$ the acceptance probability for any worsening move $\to 0$, so the algorithm degenerates into hill climbing. The risk is that it becomes trapped in whatever local optimum it happens to occupy, which is why cooling too quickly defeats the purpose of the method. **(3 marks.)**

----- End of Answer Key -----